More on Digital Signal(06-08-26)

Wednesday, August 05, 2026 11:59 AM

Baseband Signal

$$\boxed{x(n) = \sum_{k=-\infty}^{\infty} x(k)\,\delta(n-k)}$$

$x(t) = A\cos(\Omega t + \phi) \rightarrow \frac{\Omega}{T} \rightarrow x(n) = \cos(\Omega nT + \phi)$

$x(n) = \cos(w_0 n + \phi)$

$\Rightarrow w_0 = \Omega T = \frac{\Omega}{f_s} = \frac{2\pi f}{f_s} = 2\pi \frac{\Omega}{\Omega_s}$

$\frac{f}{f_s} \Big/ \frac{\Omega}{\Omega_s}$

Ω = Angular freq. $\rightarrow$ radian/sec

f = Cyclic freq.

w_0 = dimensionless = radian

$\hookrightarrow$ Normalised digital frequency

$w_0 = 2\pi \frac{f}{f_s} = \frac{2\pi\, f_s/2}{f_s} = \pi$

$\boxed{0 < w_0 < \pi}$

$0 < w_{max} < \pi$

$\boxed{-\pi < w_{max} < \pi}$

Sampling Theorem

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Aliasing effect

$\cos(\omega_0 n)$

$\omega_0 = 6\pi$

$2\pi f = 6\pi$

$f = 3$

	x_1	x_2	x_3
	$\cos(6\pi n)$	$\cos(14\pi n)$	$\cos(26\pi n)$
freq.	3	7	13

Samp fre $T_s = 0.1\text{sec}$ freq. $= 10$ hz

$\cos(0.6\pi n)$ $\cos(1.4\pi n)$ $\cos(2.6\pi n)$

$\cos(\omega_0 n)$

$= \cos\left(2\pi \frac{f}{f_s} n\right)$

$= \cos\left(2\pi \frac{3}{10}\right)$

$= \cos(.6\pi)$

$= \cos(2\pi n - 0.6\pi n)$

$= \cos(0.6\pi n)$

$= \cos(2\pi n + 0.6\pi n)$

$= \cos(0.6\pi n)$

Sampfr. $T_s = 0.02\text{sec}$, $f_s = 50\text{hz}$

$\cos(0.12\pi n)$ $\cos(0.28\pi n)$ $\cos(0.52\pi n)$

Simple and Complex system

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$$x(n) \rightarrow \boxed{DS} \rightarrow y(n) \Rightarrow \boxed{y = f(x)}$$

$$x(n) \rightarrow \boxed{f} \rightarrow y(n)$$

$$\boxed{E = y - \hat{y}}$$

$$E = 0 \Rightarrow \boxed{y = \hat{y}}$$

—— X ——

Ex 1 **Accumulator**

$$x(n) = \{\underset{\uparrow}{1}, 2, 3, 4, 5\}$$

$$\boxed{y(n) = \sum_{l=-\infty}^{n} x(l)} \rightarrow (1) \quad 1$$

$$y(n) = \sum_{l=-\infty}^{-1} x(l) + \sum_{l=0}^{n} x(l) \quad -(2)$$

$$\sum_{l=-\infty}^{n-1} x(l) + x(n)$$

$$y(n) = y(n-1) + x(n)$$

$$= y(-1) + x(n)$$

$$y(0) = y(-1) + x(0)$$

$$y(1) = y(0) + x(1) = y(-1) + x(0) + x(1)$$

Characterization of Systems

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1) Rate of sampling

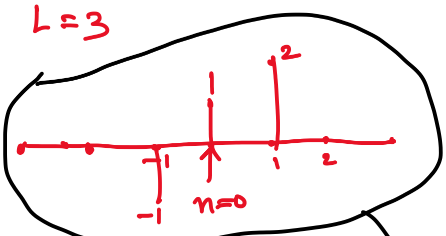
$x(n) \rightarrow \boxed{\uparrow\downarrow} \rightarrow y(n)$

Upsampling $\Rightarrow$ $y(n) = \begin{cases} x\left(\frac{n}{L}\right), & n \text{ is a int. multiple of } 0, \pm L, \pm 2L, \ldots \\ 0, & \text{otherwise} \end{cases}$

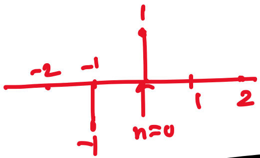
$n = 0, \pm L, \pm 2L, \ldots$

L = Rate of upsampling

$L = 3$



$x(n) = \{-1, \underset{\uparrow}{1}, 2\}$



$y(0) = x(0) = \{1\}$

$y(1) = 0$

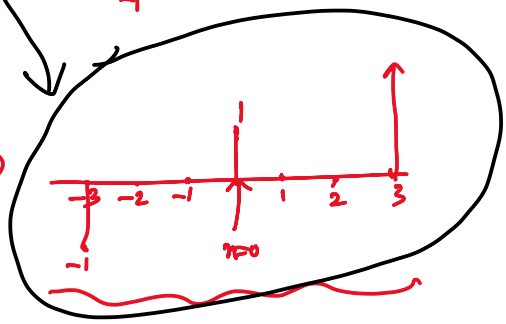
$y(2) = 0$

$y(3) = x(1) = \{2\}$

$y(-1) = 0$

$y(-2) = 0$

$y(-3) = x(-1)$



$y(n)$

— x —

Down Sampling

$x(n) \rightarrow \boxed{\downarrow M} \rightarrow y(n)$

$y(n) = \begin{cases} x(nM), & M = \pm 1, \pm 2 \\ 0, & \text{elsewhere} \end{cases}$

$M = 2$

$x(n) = \{-1, -2, \underset{\uparrow}{1}, 3, 4, 1, 5\}$

$y(n) = \{-1, \underset{\uparrow}{1}, 4, 5\}$

—— x ——

(MA)

Ex $y(n) = \frac{1}{M} \sum_{k=0}^{M-1} x(n-k)$

$$\boxed{\underline{\underline{x(n)}} = \underbrace{s(n)}_{\text{original}} + \underbrace{\underline{d(n)}}_{\text{noise}}} \approx 0$$

Linear/Nonlinear

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DS is linear iff $x_1(n) \rightarrow y_1(n) = T[x_1(n)]$

$x_2(n) \rightarrow y_2(n)$

① Homogenity

② Superposition

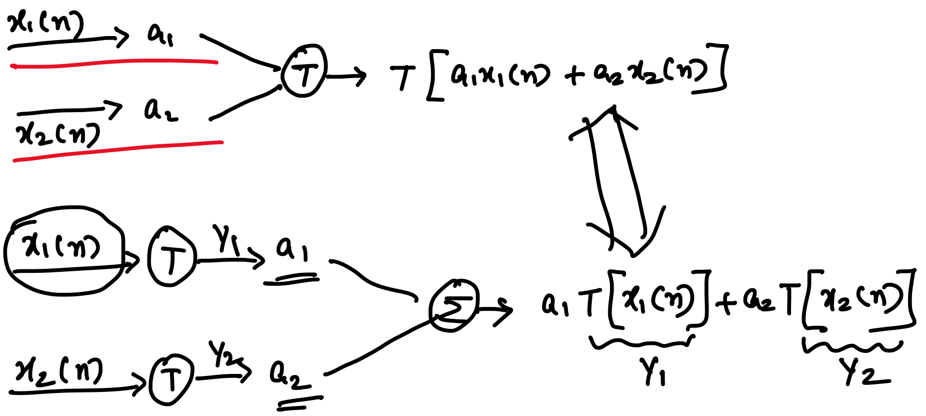
$\alpha, \beta \,/\, a_1, a_2$

$$T\left[a_1 x_1(n) + a_2 x_2(n)\right] = a_1 y_1(n) + a_2 y_2(n)$$

$$a_1 T[x_1(n)] + a_2 T[x_2(n)]$$

a_1, a_2 $\quad a_2 = 0$

$a_1 x_1(n) \rightarrow a_1 y_1(n) \rightarrow$



a_1, a_2

—— x ——

$$y(n) = \sum_{l=-\infty}^{\infty} x(l) \rightarrow$$

$\alpha x_1(n) \rightarrow \alpha y_1(n)$

$\beta x_2(n) \rightarrow \beta y_2(n)$

$\Rightarrow \alpha x_1(n) + \beta x_2(n) = \alpha y_1(n) + \beta y_2(n)$

$$\boxed{y(n) = y(-1) + \sum_{l=0}^{N-1} x(l)} \Rightarrow NL$$

$$\boxed{y(n) = n x(n)}$$

$y_1(n) = n x_1(n) = T[x_1(n)]$

$y_2(n) = n x_2(n) = T[x_2(n)]$

$a_1 y_1(n) + a_2 y_2(n) = a_1 n x_1(n) + a_2 n x_2(n)$

$$T[a_1x_1(n)+a_2x_2(n)] = n[a_1x_1(n)+a_2x_2(n)]$$
$$=[n\,a_1x_1(n)+a_2x_2(n)]$$

$$y(n) = x(n^2)$$

$$y_1(n) = x_1(n^2)$$
$$y_2(n) = x_2(n^2)$$
$$a_1y_1(n)+a_2y_2(n) = a_1x_1(n^2)+a_2x_2(n^2)$$

$$T[a_1x_1(n)+a_2x_2(n)] = a_1x_1(n^2)+a_2x_2(n^2)$$

$$\boxed{y(n) = x^2(n)}$$

$$y_1(n) = x_1^2(n)$$
$$y_2(n) = x_2^2(n)$$
$$\underline{a_1y_1(n)+a_2y_2(n)} = a_1x_1^2(n)+a_2x_2^2(n)$$
$$T[a_1x_1(n)+a_2x_2(n)] = [a_1x_1(n)+a_2x_2(n)]^2$$

$$\boxed{y(n) = x(n-1)\cdot x(n+1)} \quad ①$$

$$\underline{y(n) = x(n-1)+x(n+1)} \rightarrow ②$$

— X —

$$\boxed{y(n) = mx(n)+c}$$

$$y_1(n) = mx_1(n)+c$$
$$y_2(n) = mx_2(n)+c$$
$$a_1y_1(n)+a_2y_2(n) = a_1[mx_1(n)+c]+a_2[mx_2(n)+c]$$
$$= a_1mx_1(n)+a_1c+a_2mx_2(n)+a_2c$$

$$T[a_1x_1(n)+a_2x_2(n)] = m[a_1x_1(n)+a_2x_2(n)]+c$$
$$= ma_1x_1(n)+ma_2x_2(n)+c$$

$$\boxed{y = e^{x(n)}}$$

— x —

Time / Shift invariant

$$\boxed{x(n) \rightarrow y(n) \Rightarrow x(n-k) \rightarrow y(n-k)}$$

1) **Upsampler** $L=3$

$$\Rightarrow y(n) = \begin{cases} x\left(\frac{n}{L}\right) & n = 0, \pm L, \pm 2L \ldots \\ 0, & \text{otherwise} \end{cases}$$

$$x(n) \rightarrow x\left(\frac{n}{L}\right) \rightarrow P \rightarrow x(P)$$

$$x(n-n_0) \rightarrow x\left(\frac{n-n_0}{L}\right)$$

$$x(P-n_0) = x\left(\frac{n}{L} - n_0\right)$$

$$x(n-n_0) \rightarrow x\left(\frac{n}{L} - n_0\right)$$

$$y(n) \rightarrow y(n-n_0) = x\left(\frac{n-n_0}{L}\right)$$

— x —

Downsample

$$y(n) = \begin{cases} x(nM), & M = \pm 1, \pm 2, \ldots \\ 0, & \text{otherwise} \end{cases}$$

$$x(n) \rightarrow x(nM)$$

$$x(n-n_0) \rightarrow x[nM - n_0]$$

$$y(n) \rightarrow y(n-n_0) \rightarrow x((n-n_0)M) = x(nM - n_0M)$$

— x —

$$\boxed{y(n) = x(n-1) + x(n)} \rightarrow$$

$$\boxed{y(n) = x(-n)} \rightarrow \text{Folder}$$

TV ?

TIV ?

$$\boxed{y(n) = n\,x(n)} \rightarrow$$

Causality   Non C

$y\,[\underline{x(n)\ ,\ x(n-k)}]$ — C

$y\,[x(n+k)]$ — NC

$\boxed{y(n) = x(-n)}$

$\boxed{y(n) = x(2n)}$

$y(n) = x^2(n) + x(n-1)$
$+ x(n+1)$

Causal/Non-causal (14Aug 26)

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$$\underbrace{y(n) = f[x(n), x(n-1)]}_{C} \quad , \quad \underbrace{y(n) = f[x(n), x(n-1), x(n+1)]}_{NC}$$

$x_1(n) \equiv x_2(n)$ for $n \leq N$

Then $y_1(n) = y_2(n)$ for $n \leq N$ — **Causal**

Ex:

$y(n) = x(n) \rightarrow C$

$y(n) = x(2n) \rightarrow NC$

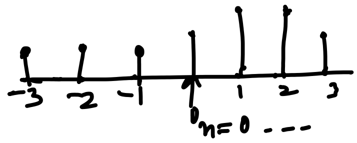
$y(n) = x(n^2) \rightarrow NC$

$\boxed{y(n) = a x(n)}$

$y(n) = x(n) + x(n+1)$

— x —

x(n), if $\boxed{x(n) = 0 , n < 0}$



$\rightarrow x(n) = \{1, 1, 1, 1.5, 2, 2, 2.5\}$

Causal Signal

— x —

Static & Dynamic

— x —

Stability **BIBO**

Stable iff $|x(n)| \leq B_x < \infty \ \forall n$

$\Rightarrow |y(n)| \leq B_y < \infty \ \forall n$

1) $y(n) = x^2(n) \rightarrow ?$

2) $\boxed{y(n) = \frac{1}{M} \sum_{k=0}^{M-1} x(n-k)}$ **BIBO**

3) **Accumulator** $y(n) = \sum_{k=0}^{\infty} x(n-k)$

$N-1$

$$\boxed{y(n) = \sum_{k=0} x(n-k)}$$ NR

$$\boxed{y(n) = \frac{y(n-1)}{y(-1)} + \sum_{k=0}^{N-1} x(n-k)}$$ ✓ — Recursive

— x —

Passivity

$$\Rightarrow \sum_{n=-\infty}^{\infty} |y(n)|^2 \leq \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$<$ $=$

→ Lossy system

→ Lossless system

$$\boxed{y(n) = \alpha x(n-N)}$$

$|\alpha| < 1 \rightarrow$ passive

$|\alpha| > 1 \rightarrow$ Active

— x —

$$\boxed{y(n) = x(-n)}$$ → Linear → ✓

Causal → NC

Time Variant →

Stable → ✓

passive → ✓

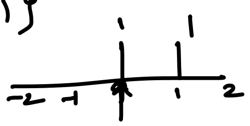
$y(n) = x(-n)$

$x(-n) \xrightarrow[\text{delayed by } k]{} x(-n-k)$

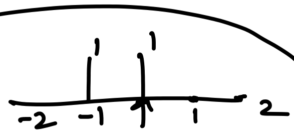
$\rightarrow y(n-k) = x(-(n-k)) = x(-n+k)$

$= \{0, 0, 1, 1\}$

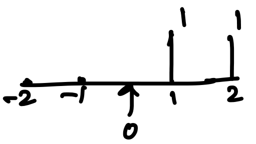
$x(n) \rightarrow$



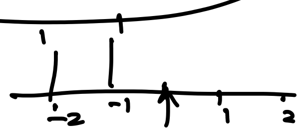
$y(n) =$



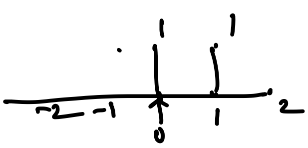
$x(n-1) =$



$\Rightarrow y(n) =$



$y(n-1) =$



Impulse and Step response

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$$\delta(n) \rightarrow \boxed{DS} \rightarrow \boxed{h(n)}$$

$$u(n) \rightarrow \boxed{DS} \rightarrow \boxed{s(n)}$$

— x —

Ex 1) **Accumulator (1)**

$$\boxed{y(n) = \sum_{l=-\infty}^{\infty} x(l)} \rightarrow h(n),\ s(n)$$

$x(l) = \delta(l)$

$$\boxed{y(n) = h(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} = u(n)}$$

$x(l) = u(l)$

$$\Rightarrow s(n) = y(n) = \begin{cases} n + \delta(n), & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$\left.\begin{array}{l} n=0,\ =1 \\ n=1,\ =1 \\ n=2,\ = \end{array}\right\} \quad ?$$

Accumulator - 2

$$\boxed{y(n) = \sum_{K=0}^{\infty} x(n-K)}$$

$h(n) \rightarrow \sum_{K=0}^{\infty} \delta(n-K) \rightarrow u(n)$

$s(n) \rightarrow$

— x —

Accumulator - 3

$$\boxed{y(n) = \underline{\underline{y(-1)}} + \sum_{l=0}^{n} x(l)}$$

$h(n) \rightarrow \{ y(-1)+1,\ n \geq 0$

$x(-1) \rightarrow$
$s(n) \rightarrow \{ \; y(-1) \;, n<0 \}$

$\rightarrow$ L, NL

— x —

Ex 4 $\Rightarrow$ **Upsampler** $\rightarrow$ $y(n) = \begin{cases} x\left(\frac{n}{L}\right), & n = 0, \pm L, \pm 2L, \ldots \\ 0, & \text{elsewhere} \end{cases}$

$$\boxed{h(n) = \delta(n)}$$

s(n) $\quad$ $x(n) = u(n)$

$$y(n) = s(n) = \{\delta(n) + \delta(n-L) + \delta(n-2L) \; ....$$

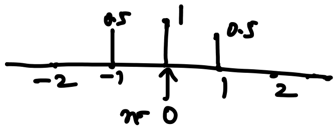
$$= \sum_{k=0}^{\infty} \delta(n-kL)$$

Downsampler find $\rightarrow$ $h(n)$, $s(n)$

Ex-6 **Interpolator**

$$\underbrace{y(n) = x(n) + \frac{1}{2}\left[x(n-1) + x(n+1)\right]}$$

$h(n) = \delta(n) + \frac{1}{2}\left[\delta(n-1) + \delta(n+1)\right] =$



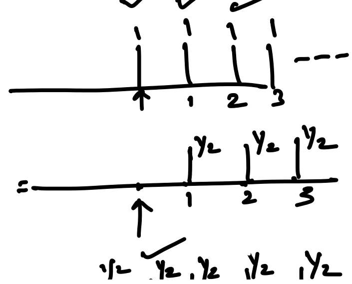
$s(n)$

$= u(n) + \frac{1}{2}\left[u(n-1) + u(n+1)\right]$

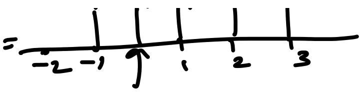
$s(n) = \{ \; , \; , \; , \; , \; , \; , \; \}$

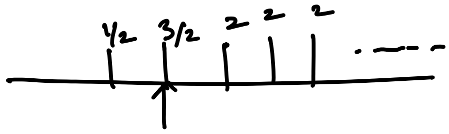
$u(n) =$ ---

$\frac{1}{2} u(n-1) =$



$s(n) = \{ 1/2, \frac{3}{2}, 2, 2, ...\}$

$\frac{1}{2}u(n+1) =$ 

$S(n) =$ 

——— x ———

LTI $\Rightarrow$ **Convolution**